

Michael Xu

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EDUCATION

University of Illinois Urbana-Champaign

Expected May 2028

Master of Science in Computer Science

University of Illinois Urbana-Champaign

May 2027

Bachelor of Science in Computer Engineering

GPA: 3.8/4.0

EXPERIENCE

Microsoft

May 2026 – Aug 2026

Software Engineering Intern

- Added snapshot API for cloning agentic sandboxes to preserve LLM session state while cutting container boot time 90%
- Contained confidential AI workloads inside Kata microVMs over userspace gVisor for agent isolation on Azure K8s Go runtime
- Designed Python data pipelines linking Copilot Shopping retrieval traces to user clicks for over 3k queries graded in Ragas
- Built MCP tools + knowledge memory layer for incident response agent with a Kusto ingestor across 3 backend stores

NVIDIA

Jan 2026 – Apr 2026

System Software Engineering Intern

- Migrated build systems to Bazel for centralizing legacy Make workflows into a Jenkins + GitLab pipeline inside a CLI app
- Cut GPU vBIOS CI build time 60% by adding caching and cross compilation to a Docker secure-boot signing pipeline
- Rewrote OTel observability platform in C++ for ingesting 2TB OTLP metrics/logs/traces using Grafana + LGTM stack
- Added LangChain hooks for tracking AI/LLM telemetry (MCP, tool calls, retrieval, agent context) into distributed traces

Northrop Grumman

May 2025 – Aug 2025

Software Engineering Intern

- Created CI/CD build pipeline with Jenkins and Kubernetes, reducing Jetson SDK service deployment times by 75%
- Reduced geospatial coordinate transformation latency 20x by using Redis caching for a stateless Flask HTTP endpoint
- Built multithreaded concurrency tests for tracking SLI/SLO using GoogleTest/pytest and raised regression coverage 91%
- Parsed real-time XML/JSON and gRPC streams converting serial I/O between microservices into InfluxDB time-series

Stanford University VISTA Lab

May 2024 – Aug 2024

Undergraduate Researcher

- Migrated plaintext credentials to AWS KMS secrets for CNN training and eval frameworks for self-driving datasets
- Rewrote JSON metadata to PostgreSQL achieving 20x faster queries for 10K+ medical imaging rows for client lookups
- Added MongoDB schema for 2K+ patient images and boosted transfer speed by 103% with concurrent Goroutines

ECE 220 - Computer Systems & Programming

Dec 2023 – Dec 2025

Undergraduate Course Assistant

- Graded C, C++, and LC-3 assembly projects focused on memory management and OOP in systems programming
- Built nightly GitLab CI job with Linux VM bash scripts and Autolab REST API tests for 5000+ assignments per semester

PROJECTS

POSIX Operating System in RISC-V | C, RISC-V Assembly, QEMU, GDB

- Built a Linux-style RISC-V kernel with virtual memory and a preemptive/cooperative round-robin scheduler exposing 30+ syscalls
- Implemented MMIO device drivers with PLIC interrupt handling backing an inode-based KTFS filesystem for persistent storage
- Designed a userspace shell with fork, pipes, redirection, and grep to spawn and manage concurrent processes

GPT-2 Forward Pass | CUDA, C++, AI/ML, cuBLAS

- Implemented forward pass for GPT-2 model layers in C/C++ with optimized Attention, LayerNorm, and FFN techniques
- Identified and resolved memory bottlenecks in Flash Attention and Brent-Kung prefix sum kernels using Nsight Systems
- Achieved 3.2x speedup for entire GPT-2 model using shared-memory tiling and KV caching with cuBLAS matmuls

SKILLS

Languages: Python, C, C++, Java, JavaScript, HTML/CSS, SQL, NoSQL, Assembly, Rust, Go (Golang), Typescript

Frameworks: React, Node.js, Pandas, Terraform, Flask, OpenCV, PyTorch, OpenTelemetry, LangChain, LangSmith, Spark

Tools: Linux, Unix, Git, Docker, Bash, AWS, Jenkins, Azure, AKS, CI/CD, Claude Code, Cursor, Prometheus, Grafana